

Problem #2: Abracadabra (15 marks)

Prof Lewos Caril, a computational linguist, has designed an artificial language *Abracadabra* that uses only five letters **A, B, C, D** and **R**. A peculiar characteristic of this language is that words are written one after another without any space or punctuation marks in between. In fact, the name of the language is actually ***Abra cad abra*** meaning “word after word”, but it is written as ***Abracadabra***. It could also be decomposed into another sequence of valid *Abracadabra* words: ***Ab rac adabra*** which translates to “the of night” and therefore, is not really a meaningful decomposition.

Consider the following Abracadabra sentence:

<i>abbaradracradababadarba</i>

Here is the dictionary of all the *Abracadabra* words along with their meanings and parts-of-speech in parenthesis (n= noun, v = verb, a = adjective or adverb, p = preposition):

<i>ab</i>	the	<i>car</i>	before (p)
<i>abra</i>	word (n)	<i>cra</i>	sun (n)
<i>adabra</i>	night (n)	<i>crada</i>	sky (n)
<i>adar</i>	teacher (n)	<i>dabab</i>	enchant (v)
<i>adrac</i>	student (n)	<i>draba</i>	dance (v)
<i>arba</i>	red (a)	<i>dara</i>	never (a)
<i>ba</i>	always (a)	<i>rac</i>	of (p)
<i>babad</i>	glow (v)	<i>racrad</i>	grow (v)
<i>bar</i>	good (a)	<i>rad</i>	blue (a)
<i>baradra</i>	morning (n)	<i>radab</i>	respect (v)
<i>cad</i>	after (p)		

Assignment 1:

Write down all possible ways in which this sentence can be decomposed into sequences of valid *Abracadabra* words.

Assignment 2:

Not all of these decompositions are grammatically correct or meaningful sentences. Which of these decompositions do you think are correct and meaningful? Give their possible English translations. Beware that Abracadabra grammar can be different from English grammar.

Prof Lewos Caril also designed a computer program to automatically segment an Abracadabra sentence, which works as follows. It scans the sentence from left to right and identifies the longest valid Abracadabra word that matches the sentence from the left. This word is segmented from the sentence, and the remaining part of the sentence is then decomposed likewise by iteratively removing the longest matching valid word from the left. At any point of time, if no matching valid word can be found, then the program stops and declares that the sentence *cannot be decomposed*. Otherwise, the program gives as output, one of the possible decompositions of the sentence.

Assignment 3:

What will be the output of Prof Caril’s program for the above sentence?

Assignment 4:

Is it possible that a sentence has a valid decomposition but Prof Caril’s program declares that it cannot be decomposed? If yes, construct such an example sentence and clearly show the valid decomposition as well as the invalid partial decomposition that Prof Caril’s program produces. If your answer is *no*, then explain why not?

In many languages, such as Thai and Chinese, words are not separated by spaces or punctuation marks. Word segmentation, the process of breaking a sentence into words, is an important challenge that a computer needs to solve before processing these types of languages.
